

Static Spherically Symmetric Space-Time

Metric:

$$ds^2 = -e^{2\Phi(r)} dt^2 + e^{2\Lambda(r)} dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2$$

Metric Tensor

$$g_{\mu\nu} = \begin{bmatrix} -e^{2\Phi(r)} & 0 & 0 & 0 \\ 0 & e^{2\Lambda(r)} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2(\theta) \end{bmatrix}$$

Christoffel Symbols

$$\Gamma_{01}^0 = \frac{d}{dr} \Phi(r)$$

$$\Gamma_{00}^1 = e^{-2(\Lambda(r))+2\Phi(r)} \frac{d}{dr} \Phi(r)$$

$$\Gamma_{11}^1 = \frac{d}{dr} (\Lambda(r))$$

$$\Gamma_{22}^1 = -r e^{-2(\Lambda(r))}$$

$$\Gamma_{33}^1 = -r e^{-2(\Lambda(r))} \sin^2(\theta)$$

$$\Gamma_{12}^2 = \frac{1}{r}$$

$$\Gamma_{33}^2 = -\frac{\sin(2\theta)}{2}$$

$$\Gamma_{13}^3 = \frac{1}{r}$$

$$\Gamma_{23}^3 = \frac{1}{\tan(\theta)}$$

Riemann Curvature Tensor

$$R_{101}^0 = \frac{d}{dr} (\Lambda(r)) \frac{d}{dr} \Phi(r) - \left(\frac{d}{dr} \Phi(r) \right)^2 - \frac{d^2}{dr^2} \Phi(r)$$

$$R_{202}^0 = -r e^{-2(\Lambda(r))} \frac{d}{dr} \Phi(r)$$

$$R_{303}^0 = -r e^{-2(\Lambda(r))} \sin^2(\theta) \frac{d}{dr} \Phi(r)$$

$$R_{001}^1 = \left(\frac{d}{dr} (\Lambda(r)) \frac{d}{dr} \Phi(r) - \left(\frac{d}{dr} \Phi(r) \right)^2 - \frac{d^2}{dr^2} \Phi(r) \right) e^{-2(\Lambda(r))+2\Phi(r)}$$

$$R_{212}^1 = r e^{-2(\Lambda(r))} \frac{d}{dr} (\Lambda(r))$$

$$R_{313}^1 = r e^{-2(\Lambda(r))} \sin^2(\theta) \frac{d}{dr} (\Lambda(r))$$

$$R_{002}^2 = -\frac{e^{-2(\Lambda(r))+2\Phi(r)} \frac{d}{dr} \Phi(r)}{r}$$

$$R_{112}^2 = -\frac{\frac{d}{dr} (\Lambda(r))}{r}$$

$$R_{323}^2 = \sin^2(\theta) - e^{-2(\Lambda(r))} \sin^2(\theta)$$

$$R_{003}^3 = -\frac{e^{-2(\Lambda(r))+2\Phi(r)} \frac{d}{dr} \Phi(r)}{r}$$

$$R_{113}^3 = -\frac{\frac{d}{dr} (\Lambda(r))}{r}$$

$$R_{223}^3 = -1 + e^{-2(\Lambda(r))}$$

Ricci Tensor

$$R_{00} = \frac{\left(-r \frac{d}{dr}(\Lambda(r)) \frac{d}{dr}\Phi(r) + r \left(\frac{d}{dr}\Phi(r)\right)^2 + r \frac{d^2}{dr^2}\Phi(r) + 2 \frac{d}{dr}\Phi(r)\right) e^{-2(\Lambda(r)) + 2\Phi(r)}}{r}$$

$$R_{11} = \frac{d}{dr}(\Lambda(r)) \frac{d}{dr}\Phi(r) - \left(\frac{d}{dr}\Phi(r)\right)^2 - \frac{d^2}{dr^2}\Phi(r) + \frac{2 \frac{d}{dr}(\Lambda(r))}{r}$$

$$R_{22} = \left(r \frac{d}{dr}(\Lambda(r)) - r \frac{d}{dr}\Phi(r) + e^{2(\Lambda(r))} - 1\right) e^{-2(\Lambda(r))}$$

$$R_{33} = \left(r \frac{d}{dr}(\Lambda(r)) - r \frac{d}{dr}\Phi(r) + e^{2(\Lambda(r))} - 1\right) e^{-2(\Lambda(r))} \sin^2(\theta)$$

Ricci Scalar

$$R = \frac{2\left(r^2 \frac{d}{dr}(\Lambda(r)) \frac{d}{dr}\Phi(r) - r^2 \left(\frac{d}{dr}\Phi(r)\right)^2 - r^2 \frac{d^2}{dr^2}\Phi(r) + 2r \frac{d}{dr}(\Lambda(r)) - 2r \frac{d}{dr}\Phi(r) + e^{2(\Lambda(r))} - 1\right) e^{-2(\Lambda(r))}}{r^2}$$

Einstein Tensor

$$G_{00} = \frac{\left(2r \frac{d}{dr}(\Lambda(r)) + e^{2(\Lambda(r))} - 1\right) e^{-2(\Lambda(r)) + 2\Phi(r)}}{r^2}$$

$$G_{11} = \frac{2r \frac{d}{dr}\Phi(r) - e^{2(\Lambda(r))} + 1}{r^2}$$

$$G_{22} = r \left(-r \frac{d}{dr}(\Lambda(r)) \frac{d}{dr}\Phi(r) + r \left(\frac{d}{dr}\Phi(r)\right)^2 + r \frac{d^2}{dr^2}\Phi(r) - \frac{d}{dr}(\Lambda(r)) + \frac{d}{dr}\Phi(r)\right) e^{-2(\Lambda(r))}$$

$$G_{33} = r \left(-r \frac{d}{dr}(\Lambda(r)) \frac{d}{dr}\Phi(r) + r \left(\frac{d}{dr}\Phi(r)\right)^2 + r \frac{d^2}{dr^2}\Phi(r) - \frac{d}{dr}(\Lambda(r)) + \frac{d}{dr}\Phi(r)\right) e^{-2(\Lambda(r))} \sin^2(\theta)$$